Cedar Site Bai

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Objective

Ph.D. cand. in CS with a strong background in machine learning, optimization, and statistics. Experienced in their applications to language model pre-training, fine-tuning, recommendation, and AI safety, with **oral presentations at ICML, ICLR, COLT** and **internships at Amazon, TikTok**. Seeking a **Researcher/Scientist** role.

Education

- **Purdue University** — Jan. 2021 – May. 2026 (Exp.)
  Ph.D. in Computer Science *(Machine Learning & Optimization)* — *Advisor: Prof. Brian Bullins*
  M.S. in Statistics & Computer Science *(along the way)* — West Lafayette, IN, US
- **Xi'an Jiaotong University (XJTU)** — Sep. 2016 – Jun. 2020
  B.S. in Computer Science *(Qian Xuesen Honors College)* — *Xi'an, China*
  Selected Awards: "Siyuan" Scholarship (twice); National 2^{nd} Prize in Mathematical Contest in Modeling (CUMCM, top 3%)

Industry Experience

Amazon, Prime Video Personalization, Applied Scientist Intern — May. 2025 – Sep. 2025
Work on **conversational recommendation** — *Mentors: Drs. Duanshun Li, Zhenyu Liao*
- Designed a reward quantifying information gain via uncertainty reduction in multi-turn conversations
- Built a data-generation pipeline integrating LLM recommenders, judges, and user simulators using real-world data
- Fine-tuned LLMs with SFT, DPO, PPO, increasing conversation efficiency by 7% and hit rate by 2%

TikTok, Shop Mall Ads, Machine Learning Engineer Intern — May. 2024 – Aug. 2024
Work on **ads recommendation ranking models** — *Mentor: Du Zhang, Manager: Dr. Yan Yan*
- Delivered two launches that improved advertiser value by 1.63% and 5.94% (**$X00,000 in profits**) after A/B test
- Applied neural architecture search to select features/emb. sizes, reducing the model size & improving accuracy
- Adapted Deep Cross Network for improved conversion rate (CVR) prediction on sequential data

Selected Publication

[1] **[ICML'25, Oral (Top 1%)] C. S. Bai***, A. Y. Zheng*, B. Bullins, R. A. Yeh. Model Immunization from a Condition Number Perspective. [PDF]

[2] **[ICML'25] C. S. Bai***, X. Luo*, B. Li*, P. Drineas, R. Zhang, B. Bullins. Stacey: Promoting Stochastic Steepest Descent via Accelerated ℓ_p-Smooth Nonconvex Optimization. [PDF]

[3] **[COLT'25] C. S. Bai**, B. Bullins. Faster Acceleration for Steepest Descent. [PDF]

[4] **[ICLR'25, Oral (Top 1.7%)] C. S. Bai**, B. Bullins. Tight Lower Bounds under Asymmetric High-Order Hölder Smoothness and Uniform Convexity. [PDF]

[5] **[IJCAI'21]** H. Zhang, **S. Bai**, X. Lan, D. Hsu, N. Zheng. Hindsight Trust Region Policy Optimization. [PDF]

Professional Techniques

Machine learning (ML): Optimizer design/analysis, reinforcement learning/RLHF, recommendation, unlearning;
Programming: Python, MATLAB (Proficient); Shell script, C/C++, R, Java, SQL, HTML;
Packages / Tools: PyTorch, NumPy, Linux Command, Git (Proficient); AWS, scikit-learn, TensorFlow, Pandas, Slurm.

Additional Publication

[1] **[Preprint] C. S. Bai**, A. Y. Zheng, R. A. Yeh, B. Bullins. Spectral Saliency for Machine Unlearning.

[2] **[Preprint]** Z. Song, **C. S. Bai**, Z. Zhang, B. Bullins, D. Gleich. Decoupling Variance and Scale-Invariant Updates in Adaptive Gradient Descent for Unified Vector and Matrix Optimization.

[3] **[Preprint]** H. Zhang, T. Hu, **C. S. Bai**, A. Xiao, D. Hsu. Uncertainty Estimation for Visual Grounding in the Open World.

[4] **[ICLR'24] S. Bai**, B. Bullins. Local Composite Saddle Point Optimization. [PDF]

[5] **[TMLR'24] S. Bai**, C. Ke, J. Honorio. On the Dual Problem of Convexified Convolutional Neural Networks. [PDF]

[6] **[IROS'19]** H. Zhang, X. Lan, **S. Bai**, X. Zhou, Z. Tian, N. Zheng. ROI-based Robotic Grasp Detection for Object Overlapping Scenes. [PDF]

[7] **[IROS'19]** H. Zhang, X. Lan, **S. Bai**, L. Wan, C. Yang, N. Zheng. A Multi-task Convolutional Neural Network for Autonomous Robotic Grasping in Object Stacking Scenes. [PDF]

Selected Courses

Machine Learning (ML): Statistical ML, ML Theory, Computer Vision, Data Mining, Artificial Intelligence
Math & Stats: Optimization, Probability, Statistical Inference, Computational Stats, Multivariate Analysis
Computer Science: Data Structures, Algorithms, Database, Machine Structure, Operating Systems, Networks

Teaching & Academic Service

Purdue CS571 Artificial Intelligence — TA (Fall 23)
Purdue CS182 Foundations of Computer Science (Discrete Mathematics) — TA (Spring 23)
Purdue CS251 Data Structures and Algorithms — TA (Fall 21, Spring 22, Fall 22)

- ICLR 2026, 2025, 2024 — Reviewer
- NeurIPS 2025, 2024, 2023 — Reviewer
- ICML 2026, 2025, 2024 — Reviewer
- AISTATS 2025, 2023 — Reviewer
- AAAI 2026, 2025 — Reviewer
- TPAMI — Reviewer
- TMLR — Reviewer